Spencer Lee

East Lansing, MI | +1 248-568-8602 | leespen1@msu.edu | spencerlee.net | github.com/leespen1

Computational-mathematics PhD researcher developing pulse-level control, calibration, and high-order simulation methods for superconducting qubits, including hands-on work on experimental qudit hardware.

Work & Research Experience

Graduate Research Assistant — **Sep 2021 - Present**
Michigan State University — *East Lansing, MI*
- Created QuantumGateDesign.jl, an open-source Julia package for pulse-level optimal control of superconducting transmon qubits via high-order Hermite time-stepping of the device Hamiltonian and discrete-adjoint gradients of the gate infidelity.
- Designing neural-network surrogate models for controlled qubit dynamics using learned Magnus expansions, enabling large-time-step state-vector simulation at low computational cost.
- Demonstrated 3–13× faster CNOT gate design than Störmer-Verlet at 10^{-3}–10^{-5} infidelity targets for transmon qubits using high-order Hermite integration and discrete adjoints (*J. Comput. Phys.*, 2026).
- Designing high-order Filon integrators for highly oscillatory quantum dynamics, enabling large, stable time-steps without the rotating wave approximation (in prep., *SIAM J. Sci. Comput.*).
- Presented talks and tutorials at international conferences, including *SIAM Annual Meeting* (2024 & 2025).

Project Manager — **Jun 2024 - Aug 2024**
Graduate-level Research in Industrial Projects for Students, Tohoku University — *Sendai, Japan*
- Implemented GPU-accelerated algorithms for classical parameter-setting of the Quantum Approximate Optimization Algorithm (QAOA) in collaboration with Mitsubishi Electric.
- Led a five-member U.S.–Japan team that replaced a state-of-the-art parameter-setting method with a low-cost analytical proxy at minimal accuracy cost.

Research Collaborator — **Sep–Dec 2023, Mar–Jul 2025**
Institute for Pure and Applied Mathematics, University of California, Los Angeles — *Los Angeles, CA*
- Organized a working group on optimal-transport-inspired objective functions for quantum control.
- Collaborated across academia and industry on quantum computing and non-commutative optimal transport.

Visiting Scholar — **Jun 2023 - Aug 2023**
Los Alamos National Laboratory — *Los Alamos, NM*
- Wrote CUDA code to accelerate simulation of runaway electrons in tokamak fusion reactors.

Computing Student Intern — **May 2022 - Aug 2022**
Lawrence Livermore National Laboratory — *Livermore, CA*
- Calibrated gate pulses and implemented zero-noise extrapolation on real superconducting qudit hardware.
- Developed high-order compositional integrators for quantum state-vector simulation.
- Demonstrated 10× lower computational cost than second-order Störmer-Verlet for 2-qubit simulations.

Education

PhD in Computational Mathematics, Science, & Engineering — **Dec 2026 (Expected)**
Michigan State University — *3.94/4.0 GPA*

Dual Bachelor of Science in Advanced Mathematics and Physics — **May 2020**
Michigan State University — *High Honors, Honors College, 3.97/4.0 GPA*

Skills

Languages	Julia, Python, C/C++, VHDL, MATLAB
HPC & Sci. Computing	CUDA, MPI, OpenMP, SLURM, adjoint methods, high-order methods
Quantum Computing	QuTiP, Qiskit, pulse-level control, QAOA, error mitigation
Machine Learning / AI	TensorFlow, Scikit-Learn, Lux.jl, scientific machine learning
Tools	Git, Linux, pprof, Claude Code (agentic AI), LaTeX

Awards & Honors

National Science Foundation Graduate Research Fellowship — **May 2022 - May 2027**
Michigan State University Rasmussen Fellowship Award — **Sep 2021**